Power Budget Example

Team Number:	307
Project Name:	Submersible Exploration Device
Team Member Names:	Zane Brauer
Version:	1

A. List ALL major components (active devices, integrated circuits, etc.) except for power sources, v

All Major Components	Component Name	Part Number	yVoltageRan	#	aximumCurre
	MicroController	PIC18F47K42	2.3 V–5.5 V	1	500
	Temperature Sensor	TMP1075DGKT	1.7-5.5V	1	0.05

B. Assign each major component above to ONE power rail below. Try to minimize the number of diff

+12V Power Rail	Component Name	Part Number	yVoltageRan	#	aximumCurre
	MicroController	PIC18F47K42	2.3 V–5.5 V	1	500
					Subtotal
					Safety Margin
					Total Current Required on +12V Rail
c1. Regulator or Source Ch	+12V regulator	LM7812	+12V - 35V	1	1000
					Total Remaining Current Available on +12V Rail

+5V Power Rail	Component Name	Part Number	yVoltageRan	#	aximumCurre
	Temperature Sensor	TMP1075DGKT	1.7-5.5V	1	0.05
	MicroController	PIC18F47K42	2.3 V–5.5 V	1	500
					Subtotal
					Safety Margin
					Total Current Required on +5V Rail
c2. Regulator or Source Ch	+5V Regulator	TPS560430X3FD	+5-35V	1	1000
					Total Remaining Current Available on +5V Rail

C. For each power rail above, select a specific voltage regulator using the same process as for majo

Regulator or Source Choice	+5V Regulator	TPS560430X3FD	+5-35V	1	1000

D. Select a specific external power source (wall supply or battery) for your system, and confirm that

External Power Source 1	Component Name	Part Number	yVoltageRan	Outpu	aximumCurre
Power Source 1 Selection	SMI24-24-V-P5	179-SMI24-24-V-F	110VAC	+24V	4000
Power Rails Connected to External Power Source 1	+12V regulator	LM7812	+12V - 35V	1	1000
	+5V Regulator	TPS560430X3FD	+5-35V	1	1000
					Total Remaining Current Available on External Power Source 1

D. Select a specific external power source (wall supply or battery) for your system, and confirm that

External Power Source 1	**Component Name**	**Part Number**	**yVoltageRan**	**Outpu**	**aximumCurre**
Power Source 1 Selection	SMI24-24-V-P5	179-SMI24-24-V-F	110VAC	+24V	4000
Power Rails Connected to External Power Source 1	+12V regulator	LM7812	+12V - 35V	1	1000
	+5V Regulator	TPS560430X3FD	+5-35V	1	1000
		Total Remaining Current Available on External Power Source 1			

Notes

External Supply Voltage should be determined by the dropout voltage for highest-voltage regulator (e.g., +1

If you have multiple units in your design (e.g., a base unit and remote unit) then you need a separate power

oltage regulators,	
alCurrent(mA)	Unit
500	mA
0.05	mA

erent power rails in	
alCurrent(mA)	Unit
500	mA
	mA
0	mA
0	mA
0	mA
500	mA
25%	
625	mA
1000	mA
375	mA
alCurrent(mA)	Unit
0.05	mA
500	mA
0	mA
0	mA
0	mA
500.05	mA
25%	
625.0625	mA
1000	mA
374.9375	mA

r component	
1000	mA

it can supply all of	
alCurrent(mA)	Unit
4000	mA
1000	mA
1000	mA
2000	mA

it can supply all of

alCurrent(mA)	Unit
4000	mA
1000	mA
1000	mA
2000	mA

4V for a +12V

budget for each unit